BondFlux: a simulation toolkit for dynamic bonds in polymer networks

`fix bond/flux` for LAMMPS — user manual and validation

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BondFlux simulates polymer networks whose bonds can form, break, and exchange. It is a single LAMMPS fix style, `fix bond/flux`, with two modes: Monte Carlo (MC) moves of topological bonds (`mc`), and the Sciortino three-body swap potential (`3body`). This manual explains the two methods, the LAMMPS commands, how to add the fix to LAMMPS, and the tests that validate it.

Introduction

Dynamic covalent and supramolecular bonds let a polymer network change its topology while it is deformed or heated. In associative networks (vitrimers) bonds exchange partners at a constant number of cross-links.[1] In dissociative networks bonds break and re-form, so their number follows the temperature. In multiple-network elastomers, a pre-stretched filler network breaks irreversibly under load while a permanent matrix keeps the material together.[2] BondFlux handles all three in a coarse-grained model such as the Kremer–Grest bead-spring model.[3] Exchange, formation and breaking can be combined with each other and with the irreversible scission of bonds through `bond_style quartic`.

Two routes are used in the literature. In the first, a dynamic bond is an explicit LAMMPS bond, and an MC step between MD steps changes the topology (Perego and Khabaz,[4] Xia et al.,[5] and others). In the second, a bond is a short-ranged pair attraction. A three-body repulsion enforces one bond per sticker and makes the bond swap barrier-free, so standard MD produces the exchange.[6] BondFlux provides both routes in the same fix (Table 1).

Monte Carlo moves of topological bonds

The atoms of the fix group are *stickers*. Each sticker has at most one dynamic bond, a LAMMPS bond of the type `btype`. Every N steps each sticker is visited once and offered one move (Fig. 1):

- a bonded sticker i with partner k breaks the bond with probability p_b (`break`), and otherwise attempts an exchange with a candidate j, chosen uniformly among the stickers within the reaction window of i: a *shift* i–k + j → i–j + k if j is free, or a *swap* i–k + j–l → i–j + k–l if j is bonded to l;
- a free sticker i forms a bond with probability p_f (`form`), with a free candidate j chosen uniformly.

Every bond that is broken *or* formed must have a length in the reaction window $[r_{\min}, r_c)$. This rule makes each move reversible

[1] D. Montarnal, M. Capelot, F. Tournilhac, and L. Leibler. Silica-like malleable materials from permanent organic networks. *Science*, 334:965, 2011. DOI: 10.1126/science.1212648

[2] E. Ducrot, Y. Chen, M. Bulters, R. P. Sijbesma, and C. Creton. Toughening elastomers with sacrificial bonds and watching them break. *Science*, 344:186, 2014. DOI: 10.1126/science.1248494

[3] K. Kremer and G. S. Grest. Dynamics of entangled linear polymer melts: a molecular-dynamics simulation. *J. Chem. Phys.*, 92:5057, 1990. DOI: 10.1063/1.458541

[4] A. Perego and F. Khabaz. Volumetric and rheological properties of vitrimers: a hybrid molecular dynamics and Monte Carlo simulation study. *Macromolecules*, 53:8406, 2020. DOI: 10.1021/acs.macromol.0c01423

[5] J. Xia, J. A. Kalow, and M. Olvera de la Cruz. Structure, dynamics, and rheology of vitrimers. *Macromolecules*, 56:8080, 2023. DOI: 10.1021/acs.macromol.3c01366; and J. Xia and M. Olvera de la Cruz. Dynamics and structure of unentangled associative polymers. *Macromolecules*, 57:8793, 2024. DOI: 10.1021/acs.macromol.4c01235

[6] F. Sciortino. Three-body potential for simulating bond swaps in molecular dynamics. *Eur. Phys. J. E*, 40:3, 2017. DOI: 10.1140/epje/i2017-11496-5

	`mc`	`3body`
dynamic bond topology	LAMMPS bond of one type	pair interaction within r_c
changes by	Metropolis MC every N steps	MD (continuous forces)
moves	shift, swap, formation, breaking	swap, thermal breaking and formation
number of bonds	conserved, or fluctuating with `form/break`	fluctuating (fixed if $\epsilon \gg kT$)
rate control	attempt gate $e^{-E_a/kT}$, frequency	λ, temperature

Table 1: The two modes of `fix bond/flux`.

from the same sticker with the same proposal probability: in a shift, the free stickers near i before the move are the free stickers near i after it, with j and k exchanged, and both are inside the window. Candidates must be bondable (`bondable` types, `molecule` rule) and must not already be bonded to i through another bond.

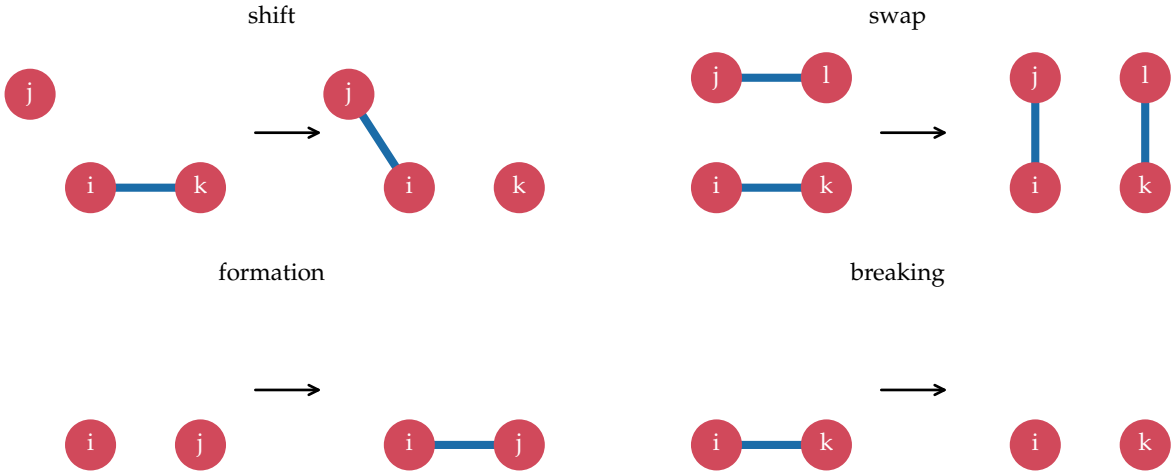


Figure 1: Monte Carlo moves of `fix bond/flux mc`: sticker i is the sticker offered the move; dynamic bonds are drawn as lines. Top: exchange (shift, swap), which keeps the number of dynamic bonds. Bottom: formation and breaking, which change it.

Energy change. The energy change of a move is evaluated with the bond and pair styles of the simulation:

$$\begin{aligned}\Delta E = &\sum_{\text{formed}} \left[U_{\text{bond}}(r) - \Delta U_{\text{pair}}(r)\right] \\ &- \sum_{\text{broken}} \left[U_{\text{bond}}(r) - \Delta U_{\text{pair}}(r)\right] \\ &+ E_{\text{bond}}\,(n_{\text{formed}} - n_{\text{broken}}),\end{aligned} \tag{1}$$

where $\Delta U_{\text{pair}}(r) = U_{\text{pair}}(r;1) - U_{\text{pair}}(r;w_{12})$ is the change of the pair energy when a pair stops being a 1–2 neighbor (w_{12} = `special_bonds` weight) and E_{bond} is an optional association energy (`ebond`). For a Kremer–Grest model with `special_bonds lj 0 1 1`, a FENE bond and a WCA pair, the WCA terms cancel and ΔE is the change of the FENE energy; with `special_bonds 1 1 1` (required by `bond_style quartic`) $\Delta U_{\text{pair}} = 0$. ΔE is exactly the change of the LAMMPS potential energy, as the validity tests confirm.

Acceptance. Exchange moves are accepted with $\min(1, e^{-\Delta E/kT})$. Formation and breaking are each other's reverse but are proposed with different probabilities (a free sticker picks one of n_c candidates; a bonded sticker breaks with probability p_b), so they carry the Metropolis–Hastings ratio

$$\begin{aligned} P_{\text{form}} &= \min\Big(1, \frac{p_b\, n_c}{p_f}\, e^{-\Delta E/kT}\Big), \\ P_{\text{break}} &= \min\Big(1, \frac{p_f}{p_b\, n_c'}\, e^{-\Delta E/kT}\Big), \end{aligned} \tag{2}$$

where n_c' is the number of free candidates of i after the bond is broken.

Barrier. A reaction barrier E_a (`barrier`) and an attempt probability P (`prob`) enter as a gate: a sticker is offered a move only with probability $g = P\, e^{-E_a/kT}$. The gate multiplies the rate of every move and of its reverse by the same factor, so it changes the kinetics and leaves the equilibrium unchanged (tested below). Adding E_a to ΔE inside the Metropolis exponent, as Xia and Olvera de la Cruz do,[7] does not have this property.

[7] J. Xia and M. Olvera de la Cruz. Dynamics and structure of unentangled associative polymers. *Macromolecules*, 57:8793, 2024. DOI: 10.1021/acs.macromol.4c01235

Parallel moves. A move centered on sticker i involves atoms within $2r_c$ of i. Each MPI sub-domain is split into 2^d sectors, and the sectors are processed one after the other; moves in the same sector of neighboring processors are then at least half a sub-domain apart and cannot conflict. Partner changes of ghost atoms are sent to their owners after each sector, also through the multi-hop paths near sub-domain corners. At the end of an MC step the bond and 1–2 special lists are updated and a reneighboring is triggered. The partner bookkeeping is rebuilt from the bond list at every MC step, so bonds broken by `bond_style quartic` are picked up automatically.

Three-body swap potential

In `3body` mode the fix adds the Sciortino potential between bondable stickers. The two-body bond is a $2n$–n Lennard-Jones potential,

$$\begin{aligned} V(r) &= \epsilon\, \frac{\phi(r) - \phi(r_c)}{1 + \phi(r_c)}, \qquad r < r_c, \\ \phi(r) &= 4\Big[\Big(\frac{\sigma}{r}\Big)^{2n} - \Big(\frac{\sigma}{r}\Big)^{n}\Big], \end{aligned} \tag{3}$$

shifted to zero at r_c and rescaled to a depth of $-\epsilon$ at $r_{\min} = 2^{1/n}\sigma$. With `eshift no`, $V = \epsilon\,\phi(r)$ is used unshifted, as in Sciortino's paper. The three-body term,

$$\begin{aligned} E_3 &= \lambda\, \epsilon \sum_i \sum_{j<k} V_3(r_{ij})\, V_3(r_{ik}), \\ V_3(r) &= \begin{cases} 1 & r \le r_{\min} \\ -V(r)/\epsilon & r_{\min} < r < r_c \end{cases}, \end{aligned} \tag{4}$$

penalizes a sticker with two partners. For $\lambda = 1$ the energy gained by a second partner is compensated exactly, so the swap is barrier-free; $\lambda > 1$ adds an exchange barrier of $(\lambda - 1)\epsilon$. Forces, energy and virial are computed in the `pre_reverse` stage on a full neighbor list, and forces on ghost atoms are summed by LAMMPS's reverse communication. The fix supplies the complete interaction between bondable sticker types; the pair style interaction between them must be switched off (`pair_coeff A B 0.0 ...`).

The three-body potential is widely used for coarse-grained vitrimers, for example by Liu, Ganesan and co-workers to relate the bond swap barrier to the structure, mechanics, self-healing and reprocessing of vitrimer networks,[8] and by Sing and co-workers to study penetrant diffusion in vitrimer networks.[9]

[8] H. Zhao, X. Wei, Y. Fang, K. Gao, T. Yue, L. Zhang, V. Ganesan, F. Meng, and J. Liu. Molecular dynamics simulation of the structural, mechanical, and reprocessing properties of vitrimers based on a dynamic covalent polymer network. *Macromolecules*, 55:1091, 2022. DOI: 10.1021/acs.macromol.1c02034

[9] M.-H. Lin, T.-W. Lin, and C. E. Sing. Molecular simulation study of penetrant diffusion in vitrimer networks. *J. Chem. Phys.*, 164:144908, 2026. DOI: 10.1063/5.0322457

Using fix bond/flux in LAMMPS

Syntax

```
fix ID group-ID bond/flux mc Nevery btype rc T seed &
    keyword value ...
fix ID group-ID bond/flux 3body eps sigma n rc lambda &
    keyword value ...
```

- `mc`: `Nevery` = steps between MC steps; `btype` = bond type of the dynamic bonds; `rc` = upper end of the reaction window (distance); `T` = temperature of the Metropolis test (use the thermostat set point); `seed` = random number seed.
- `3body`: `eps`, `sigma`, `n` = depth, size and exponent of the $2n$–n bond; `rc` = cutoff; `lambda` = three-body strength.

Table 2: Keywords.

keyword	mode	value	default
`bondable`	both	`itype jtype` (ranges allowed, repeatable)	all pairs
`shift`, `swap`	mc	`yes`/`no`	`yes`
`form`, `break`	mc	attempt probabilities p_f, p_b (both > 0 or both 0)	0, 0
`ebond`	mc	association energy per dynamic bond	0
`barrier`	mc	activation energy E_a	0
`prob`	mc	attempt probability P	1
`rmin`	mc	lower end of the reaction window	0
`molecule`	mc	`any`, `inter`, `intra`	`any`
`rbond`	3body	distance that defines a bond for the output	`rc`
`eshift`	3body	`yes`/`no`: shift V to zero at `rc`	`yes`

Output. `mc`: global vector with 11 elements (attempts, proposals, accepted shifts, swaps, formations, breakings, geometric rejections, Metropolis rejections, sum of ΔE of accepted moves, number of dynamic bonds, accepted moves in the last MC step); global scalar $E_{\text{bond}} \times$(number of dynamic bonds) when `ebond` is set, included in the potential energy. `3body`: global scalar = $E_2 + E_3$ (included in the

potential energy and virial), vector (E_2, E_3, bonded pairs, stickers with more than one partner). Per-atom array, both modes: column 1 = partner atom ID (nearest bonded partner in `3body`), column 2 = number of moves of the atom (`mc`) or number of partners within `rbond` (`3body`).

Restrictions. `mc`: molecular atom style; orthogonal box; `special_bonds lj/coul` w_{12} `1.0 1.0` (1–3 and 1–4 neighbors must not be special); `comm_modify cutoff` $\geq 2r_c + \text{skin}/2$; processor sub-domains $\geq 8r_c + \text{skin}$; `extra/bond/per/atom` and `extra/special/per/atom` when bonds can form; angles or dihedrals must not contain dynamic bonds. For FENE bonds $r_c < 0.95R_0$; for quartic bonds r_c must be shorter than the breaking distance. LAMMPS cannot combine `quartic` with unbreakable styles in `bond_style hybrid`, so a network with breakable quartic strands uses quartic dynamic bonds as well. `3body`: `newton on`; `comm_modify cutoff` $\geq r_c + \text{skin}$. Neither mode supports `run_style respa`. The fix writes no restart data: after `read_restart` it is defined again, the partners are rebuilt from the bond list, and the per-atom move counts and random number streams start anew.

Examples

Kremer–Grest vitrimer (exchange only, number of bonds conserved; `examples/kg_network`):

```
special_bonds lj/coul 0.0 1.0 1.0
bond_style    fene
bond_coeff    * 30.0 1.5 1.0 1.0
group         stick type 2
fix           flux stick bond/flux mc 100 2 1.3 1.0 777 &
              barrier 2.0
```

Dissociative network (reversible formation and breaking with association energy):

```
fix           flux stick bond/flux mc 100 2 1.3 1.0 12347 &
              form 1.0 break 0.5 ebond -24.0
```

Multiple network with breakable strands (quartic bonds; type 1 = permanent strands, type 2 = dynamic junctions; strands break irreversibly above 1.3σ):

```
special_bonds lj/coul 1.0 1.0 1.0
bond_style    quartic
bond_coeff    1 1200.0 -0.55 0.25 1.3 34.6878
bond_coeff    2 1200.0 -0.55 0.25 1.3 34.6878
fix           flux stick bond/flux mc 10 2 1.2 1.0 4321 &
              barrier 3.0
```

Cross-linker vitrimer after Xia et al.[10] (A stickers on chains, B ends of three-bead cross-linkers, LJ with $r_c = 1.5\sigma$, MC every 10 steps,

[10] J. Xia, J. A. Kalow, and M. Olvera de la Cruz. Structure, dynamics, and rheology of vitrimers. *Macromolecules*, 56:8080, 2023. DOI: 10.1021/acs.macromol.3c01366

reaction window $0.87 \le r < 1.14$; only free A stickers attack bonded B):

```
special_bonds lj/coul 0.0 1.0 1.0
bond_coeff    2 30.0 1.5 0.0 1.0      # FENE without the WCA term
fix           flux AB bond/flux mc 10 2 1.14 1.0 9876 &
              bondable 1 2 swap no rmin 0.87 barrier ${Ub}
```

Pair-swap vitrimer after Perego and Khabaz (all stickers bonded, swaps only, an MC step every MD step):

```
fix           flux stick bond/flux mc 1 2 1.3 1.0 5555 shift no
```

Sciortino dumbbells (`examples/dumbbells`; A–A and B–B repulsive $2n$–n WCA):

```
pair_style    mie/cut $(2.0^(1.0/100))
pair_coeff    * * 1.0 1.0 200 100
pair_coeff    1 2 0.0 1.0 200 100
pair_modify   shift yes
fix           tb stick bond/flux 3body 1.0 1.0 100 1.3 1.0 &
              bondable 1 2
```

The model of Lin et al. corresponds to `3body 30.0 1.0 100 1.2 lambda`.

Adding BondFlux to LAMMPS

The fix uses only core LAMMPS classes and is compiled as part of the LAMMPS core. With CMake:

```
cd lammps/src
# link (or copy) the two files
ln -s ../../src/fix_bond_flux.h ../../src/fix_bond_flux.cpp .
cd ..
cmake -S cmake -B build -D PKG_MOLECULE=on -D PKG_EXTRA-PAIR=on \
      -D BUILD_MPI=on -D BUILD_SHARED_LIBS=on
cmake --build build -j 8
build/lmp -h | grep bond/flux
```

`PKG_MOLECULE` provides the bond styles, `PKG_EXTRA-PAIR` the `mie/cut` pair style of the dumbbell examples, and `BUILD_SHARED_LIBS` the library for the Python tests. With the legacy make build, copy the two files into `lammps/src` and run `make mpi`. BondFlux was developed and tested with LAMMPS stable 30 Sep 2026.

Validity tests

The scripts are in `tests/` and `examples/`; the figure data are in `doc/data/bondflux_validation.h5`. Times are in Lennard-Jones units $\tau = \sigma\sqrt{m/\epsilon}$. The bond autocorrelation $C(t)$ is the fraction of bonds present at time 0 that are present at time t; τ_x is its decay time.

Exact detailed balance of the MC moves

Six stickers are frozen in place (no time integration), so only the MC moves change the system. The visited topologies must then follow $e^{-E/kT}$ over all topologies the moves can reach. The reachable topologies and their energies are enumerated independently of LAMMPS (`tests/test_mc_balance.py`). The test was run for four force fields: Kremer–Grest FENE with `special_bonds 0 1 1`; quartic with `1 1 1`; FENE without WCA, where bonded pairs lose their repulsion; and harmonic with a 1–2 weight of 0.5. It covered exchange only, formation and breaking only, and both together, with and without a barrier, on 1, 2 and 4 MPI ranks, with the cluster in the middle of the box and across a periodic box corner. All 48 runs pass: the largest block z-score is below 3.1, the total variation distance below 0.006, and the LAMMPS potential energy changes by the sum of the ΔE of the accepted moves to 10^{-10}. Figure 2 shows the run with formation and breaking on 4 ranks.

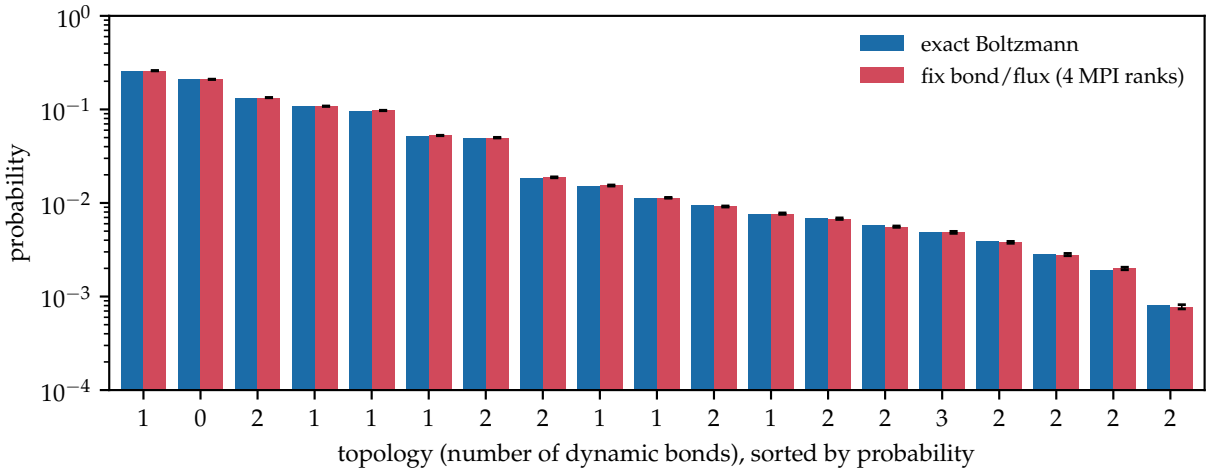


Figure 2: Probabilities of the 19 reachable topologies of a frozen six-sticker cluster (0, 1 or 2 dynamic bonds) with exchange, formation and breaking ($p_f = p_b = 0.5$, $E_{\mathrm{bond}} = -20\epsilon$, $kT = 4\epsilon$, Kremer–Grest FENE, 4×10^5 MC steps on 4 MPI ranks) compared with the exact Boltzmann distribution. Total variation distance 0.0018.

Barrier as a gate versus inside the Metropolis exponent

For the same cluster, the transition matrix of the exchange MC was built exactly and its stationary distribution compared with the Boltzmann distribution (`tests/compare_acceptance_rules.py`). With the gate of `fix bond/flux` the stationary distribution is Boltzmann for every barrier. With the barrier added to ΔE, a move and its reverse are both penalized, so for $|\Delta E| < E_a$ their acceptance ratio becomes $e^{-2\Delta E/kT}$. The equilibrium of the topology is then that of half the temperature (Table 3).

rule	window	$E_a/kT = 0$	1	2	4	8
gate	$[0, 1.3)$	2×10^{-15}	2×10^{-14}	1×10^{-14}	1×10^{-13}	2×10^{-11}
exponent	$[0, 1.3)$	2×10^{-15}	0.235	0.275	0.283	0.283
gate	$[0.87, 1.14)$	3×10^{-17}	6×10^{-16}	2×10^{-15}	1×10^{-14}	4×10^{-13}
exponent	$[0.87, 1.14)$	3×10^{-17}	0.149	0.149	0.149	0.149

Table 3: Total variation distance between the stationary distribution and the Boltzmann distribution (exact, no sampling). Gate (`fix bond/flux`): $P_{\mathrm{acc}} = e^{-E_a/kT} \min(1, e^{-\Delta E/kT})$; exponent: $P_{\mathrm{acc}} = \min(1, e^{-(\Delta E + E_a)/kT})$.

A reaction window checked only for the new bond, as in the model of Xia et al., is not reversible either: a bond stretched beyond the window can be exchanged away but never formed. This bias acts through bond-length fluctuations and does not appear for a frozen cluster.

Kremer–Grest sticker network

A melt of 200 Kremer–Grest chains of 20 beads with stickers at beads 2, 6, 10, 14 and 18 (`examples/kg_network`) is first turned into a dissociative network: stickers form and break bonds with $E_{\mathrm{bond}} = -24\epsilon$ at $kT = 1$, and the number of bonds settles at about 400 of 500 possible (Fig. 3a). With formation and breaking switched off, the number of bonds is conserved (409 bonds before and after 2×10^5 steps for every barrier) and the shift and swap moves relax the topology. The barrier enters as the gate $e^{-E_a/kT}$: going from $E_a = 0$ to 2ϵ and 4ϵ reduces the number of attempts by factors of 7.40 and 7.33 per step of $2kT$ and the number of accepted exchanges by 7.45 and 7.2, close to $e^2 = 7.39$. The bond autocorrelation time grows from $\tau_x = 128\tau$ to 739τ and about 4300τ (the last value extrapolated beyond the run), a factor of 5–6 per $2kT$, because exchanges that swap a bond back restore the original pairing (Fig. 3b).

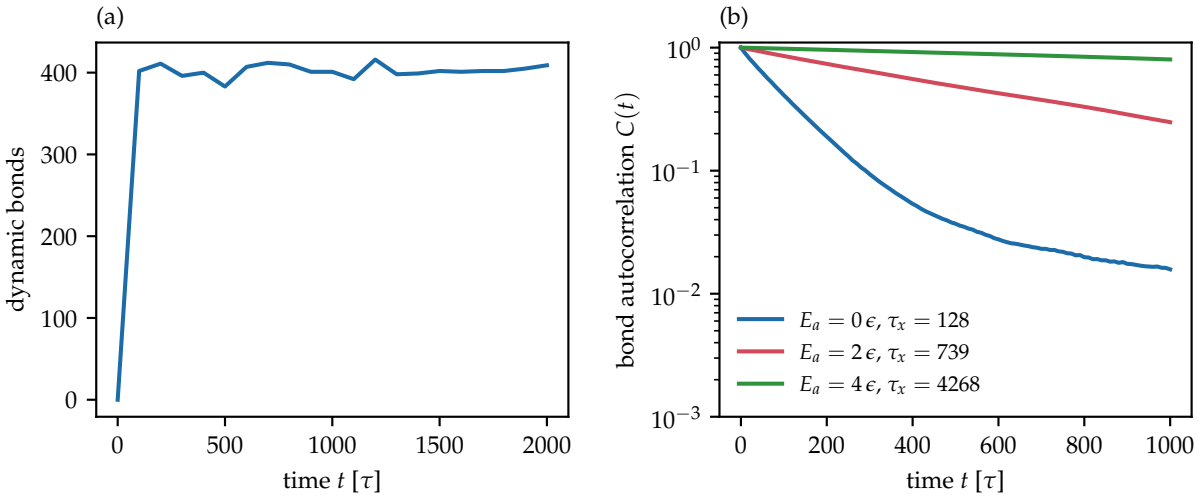


Figure 3: Kremer–Grest sticker network. (a) Number of dynamic bonds during reversible formation and breaking. (b) Bond autocorrelation $C(t)$ in the same network with exchange only, for three barriers.

Three-body forces and virial

`tests/test_3body.py` compares the forces with central finite differences of the total energy, and the virial with finite-difference strain derivatives (box scaling in x, y, z and an xy shear). The test uses $n = 12$ and 100, $\lambda = 1$ and 1.3, and 1, 2 and 4 MPI ranks. The relative errors are $\leq 8 \times 10^{-10}$ for the forces and $\leq 3 \times 10^{-9}$ for the virial, and the parallel forces agree with the serial ones to 2×10^{-13}. For an A–B bond at r_{min} and a second B approaching

from the other side, the total energy stays $-\epsilon$ to 10^{-12} while the two-body energy varies between -2ϵ and $-\epsilon$ ($n = 100$, $\lambda = 1$).

Dumbbells: comparison to the reference and bond-swap tuning

Sciortino's dumbbell system[11] has 400 A and 600 B particles at $\rho = 0.125$, with $n = 100$, $r_c = 1.3$ and $kT = 0.03$ (`examples/dumbbells/in.dumbbells`: 1000τ equilibration, 2000τ production, Langevin damping time 100). All 400 A particles are bonded at all times, and for $\lambda = 1$ the bond autocorrelation decays exponentially with $\tau_x = 116.7\tau$, against 116τ in the reference (Fig. 4a). For $\lambda = 1.1$ the lifetime grows 3.5 times to $\tau_x = 404\tau$, and for $\lambda = 1.3$ the bonds practically do not exchange within 1000τ.

[11] F. Sciortino. Three-body potential for simulating bond swaps in molecular dynamics. *Eur. Phys. J. E*, 40:3, 2017. DOI: 10.1140/epje/i2017-11496-5

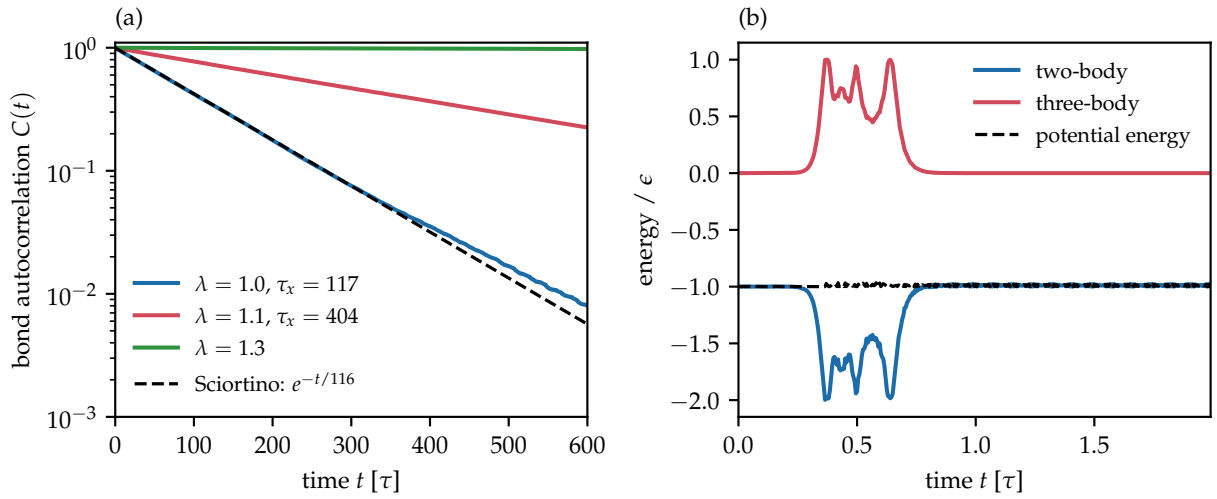


Figure 4: Three-body swap potential. (a) Bond autocorrelation $C(t)$ of the dumbbell system for $\lambda = 1$, 1.1 and 1.3, with the reference decay $e^{-t/116}$ of Sciortino. (b) Energies during a burst of bond swaps of one A and two B particles ($n = 150$, $\lambda = 1$, NVE, $kT \approx 0.01$); the two-body and three-body energies compensate and the total potential energy stays at $-\epsilon$.

Potential energy during a swap event

One A and two B particles move in NVE in a box of side 2.6 ($n = 150$, $\lambda = 1$, `examples/dumbbells/in.swap_event`). During a swap, the two-body energy drops to -2ϵ while the three-body energy rises to $+\epsilon$, so the potential energy stays at $-\epsilon$ (Fig. 4b). The total energy is conserved to 10^{-5} over 1.5×10^5 steps.

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J. Xia and M. Olvera de la Cruz. Dynamics and structure of unentangled associative polymers. *Macromolecules*, 57:8793, 2024. DOI: 10.1021/acs.macromol.4c01235.

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